Sanchit Singh

Los Angeles, CA — sanchittech0.01@gmail.com — (559) 779-2212
sanchitsingh.com — github.com/sanchitsingh001 — linkedin.com/in/sanchitsingh001

Education

San Diego State University Aug 2022 - May 2026
Bachelor of Science in Computer Science GPA: 3.57/4.00 (Major GPA: 3.84/4.00)
Dean's List (Fall 2023, Spring 2024, Fall 2024, Spring 2025, Fall 2025, Spring 2026)
Relevant Coursework: Data Structures & Algorithms, Machine Learning, Artificial Intelligence, Computer Architecture, Advanced Programming Languages, Software Systems, Probability/Statistics, Linear Algebra, Calculus, Physics

Awards & Honors

- **Honorable Mention, CRA Outstanding Undergraduate Researcher Award (2026)**.
- Awarded a **$3,000 Seed Fund (2025)** from the Lavin Entrepreneurship Center to support research and development of an independent AI venture.
- **Dean's List (2023–Present)**, San Diego State University — awarded consecutively each semester for academic excellence.
- **Finalist (2025)**, UC Berkeley AI Hackathon — 1 of 9 finalists (top ≈2.6%) among 350+ projects and 1,400+ participants.
- **TensorFlow Developer Certificate (Mar 2024)**.

Technical Skills

Machine Learning: PyTorch, TensorFlow, Hugging Face, Transformers, Vision Transformers, LLMs, VLMs, Computer Vision, Multimodal Learning, Self-Supervised Learning, Representation Learning, OOD Detection, Domain Generalization, Model Evaluation
Systems and Data: Python, C++, Docker, Linux, AWS, GCP, Weights & Biases, REST APIs, Large-Scale Data Processing

Publications

- Aditi Naiknaware*, **Sanchit Singh***, Hajar Homayouni, Salimeh Sekeh. *Temp-SCONE: A Novel Out-of-Distribution Detection and Domain Generalization Framework for Wild Data with Temporal Shift.* NeurIPS 2025 Workshop on Reliable ML for Unreliable Data **(Paper + Poster Presentation, Accepted)**.
- **Sanchit Singh**. *Training-Free Construction of Executable 3D Worlds from Narrative Text.* The 2nd Workshop on World Models: Understanding, Modelling, and Scaling, ICLR 2026 Workshop **(Accepted)**.
- Elkins, A. C., **Singh, S.***, Pourebadi, M., Amadasun, U. P., Abhari, K. *Socially Intelligent Robots with Large Behavior Models: Challenges, Strategies, and Future Research Opportunities.* International Conference on Information Systems (ICIS) 2025 **(Accepted)**.
- Elkins, A. C., **Singh, S.***, Pourebadi, M., Amadasun, U. P., Abhari, K. *Designing Socially Grounded Data Pipelines for Training and Operating Socially Intelligent Robots: Challenges and Future Directions.* Hawaii International Conference on System Sciences (HICSS) 2026 **(Accepted)**.
- Elkins, A. C., **Singh, S.***, Blankenship, S., Kieback, A., Amadasun, U. P., Chadha, A. *DAWZY: A New Addition to AI powered "Human in the Loop" Music Co-creation.* NeurIPS 2025 Workshop on AI for Music **(Paper + Demo Track + Poster Presentation, Accepted)**. *Demo Track – Extended Abstract.*

(denotes equal contribution)*

Research Experience

Human–AI Interaction in Agentic Research Systems, UCLA *Visiting Researcher* Jul 2026–Present
Advisor: Prof. Xiang "Anthony" Chen

- Conducting research on human–AI interaction in agentic research systems for scientific discovery.
- Designing and evaluating human-in-the-loop supervision mechanisms that enable AI research agents to incorporate expert feedback through structured and traceable revision workflows.

Socially Intelligent Robots (Multimodal Understanding), James Silberrad Brown Center for Artificial Intelligence at SDSU *Lead Student Researcher* Aug 2024–May 2025
Advisors: Prof. Mary Pourebadi, Prof. Kaveh Abhari, Prof. Aaron Elkins

- Designed and conducted video question-answering experiments using VLMs (LLaVA) and LLMs (LLaMA) to improve temporal scene understanding, integrating a face recognition module for identity tracking and context-aware response generation.
- Contributed to advancing the lab's vision of enabling socially aware, human-centered robots capable of understanding and responding to people in real-world environments such as hospitals and mental health settings

Out-of-Distribution Robustness (Temp-SCONE), Sekeh Lab, SDSU *Student Researcher* Feb 2025–Sep 2025
Advisors: Prof. Hajar Homayouni, Prof. Salimeh Sekeh

- Designed and evaluated a Novel Temp-SCONE framework, a temporal extension of SCONE and WOODS for robust OOD detection under evolving distributions.
- Built pipelines with WideResNet and Vision Transformers, training across temporally split datasets (CIFAR-10, CINIC-10, Imagenette)
- Contributed to theoretical analysis of temporal consistency and its relationship to generalization error under domain shifts

DAWZY: Creative AI for Human-AI Collaboration, Independent Research *Independent Research Lead & Equal First Author* Jun 2025–Sep 2025

- Independently conceived and led a research initiative exploring human–AI co-creation in digital audio workstations, focusing on feasibility, design, and the cognitive dimensions of assistive creativity.
- Designed the research and technical framework, implemented experimental prototypes, and conducted user studies to evaluate learning, usability, and collaboration in human-centered creative AI systems.

Privacy in Graph Learning, Privacy and Anonymity Lab (PAL), SDSU *Student Researcher* Jun 2025–Dec 2025
Advisor: Prof. Joann Chen

- Researched node-, edge-, and subgraph-level membership inference attacks on provenance graph datasets using Graph Neural Networks
- Developed unified pipelines with shadow/target splits to evaluate inference attacks across graph granularities

Bioacoustics Representation Learning, MAR Lab, SDSU *Student Researcher* Sep 2025–Present
Advisors: Prof. Hajar Homayouni, Prof. Marie Roch (with San Diego Zoo Wildlife Alliance)

- Applying self-supervised learning to 4 TB of rainforest audio data to cluster and analyze animal vocalizations across species and habitats.
- Developing transformer-based embedding pipelines and attention-guided clustering to detect rare and frequency-specific acoustic events.

Selected Projects

DAWZY — AI Assistant for Music *Project Lead and Developer* 2025

- Developed an AI-powered Digital Audio Workstation that enables beginners to create and edit music through natural language interaction.
- Finalist (1 of 3 in Creative AI Track; 3 tracks total, 9 finalists overall, ≈2.6% of 350+ projects) at UC Berkeley AI Hackathon 2025, selected from 1,400+ participants.
- Pitch: YouTube — Submission: Devpost.

Executable 3D Worlds from Narrative Text 2026

- Built a training-free pipeline that converts narrative text into a navigable 3D world in Godot using pre-trained LLMs and text-to-3D APIs. Code: GitHub — Submission: Devpost.